
CMS Physics Analysis Summary

Contact: cms-pag-conveners-smp@cern.ch

2025/09/30

Measurement of spin correlations induced by gluon polarization in parton showers

The CMS Collaboration

Abstract

A study of angular correlations induced by gluon polarization effects inside jets is performed using proton-proton collisions at a center-of-mass energy of $\sqrt{s} = 13.6$ TeV. The data, corresponding to an integrated luminosity of 34.7 fb^{-1} , were collected in 2022 with the CMS detector at the LHC. The details of the parton shower are studied using jets reconstructed with the anti- k_T algorithm and declustered with the Cambridge–Aachen algorithm. A new technique is developed to identify features in jet substructure and to select intermediate gluon splittings into quark-antiquark pairs. An observable sensitive to the gluon polarization effect is compared with PYTHIA 8 and HERWIG 7 parton-shower predictions, with and without spin correlations. The data confirm model predictions that include gluon polarization effects, while strongly disfavoring models without it.

The gluon is the gauge boson that mediates the strong interaction, theoretically described by quantum chromodynamics (QCD). Discovered by the TASSO [1] and PLUTO [2] experiments, its spin-1 nature has been thoroughly established in studies of the spin correlations among jets produced in e^+e^- annihilation at LEP [3, 4]. Such correlations arise from the two possible polarization states of the gluon, yielding quantum interference effects that are reflected in final-state spin-sensitive observables. However, such effects have never been directly observed in collinear and soft parton radiation inside the jets themselves, where the physics is usually described by parton shower (PS) models and affected by nonperturbative effects. Jets produced in multi-TeV proton-proton (pp) collisions at the CERN LHC have much larger energies than previously reached at e^+e^- colliders. This leads to a large dynamic range for intrajet parton radiation, and makes it possible to study spin-correlation effects in the soft and collinear regime. Most current PS Monte Carlo (MC) event generators incorporate gluon polarization effects following the original Collins–Knowles approach [5, 6]. In the PYTHIA [7] and SHERPA [8] models, a modulation is added to the opening angle between the production and splitting planes of the gluon to approximate such quantum interference effects. In a recent HERWIG [9] development, spin correlations are treated explicitly using a spin-density matrix formalism. Properly describing gluon spin effects is crucial in modern PS algorithms that go beyond the leading-logarithmic accuracy [10–15]. Moreover, jet-flavor tagging algorithms based on simulated data rely on a realistic description of the kinematics of the particles inside jets as input. Any mis-modeling of the correlation among particles, e.g. in the Higgs boson decay into two gluons, impacts the jet tagging performance. In addition, the impact of nonperturbative effects on the correlation is unknown. It is thus crucial to experimentally measure the spin correlation of particles inside jets, check their modeling in current MC generators, and provide new inputs for their improvement.

In two successive parton splittings inside a shower, $X \rightarrow 1 + 2$ followed by $2 \rightarrow 3 + 4$, the matrix elements of the helicity states of the intermediate parton 2 interfere and give rise to a nonuniform contribution to spin-sensitive observables. One such observable is the angle $\Delta\varphi$, defined as [9, 10]

$$\cos \Delta\varphi = \hat{n}_{1,2} \cdot \hat{n}_{3,4} \cdot \text{sign}(\hat{n}_{1,2} \times \hat{n}_{3,4} \cdot \vec{p}), \quad (1)$$

where $\hat{n}_{1,2}$ and $\hat{n}_{3,4}$ are unit vectors of the plane formed by partons 1, 2 and partons 3, 4, respectively, and $\vec{p}$ is the momentum of the most energetic of partons 1 and 2. The second $2 \rightarrow 3 + 4$ branching can be $g \rightarrow gg$, $g \rightarrow q\bar{q}$ and $q \rightarrow gq$. Theoretical calculations [9] predict that spin correlations only exist in $g \rightarrow gg$ and $g \rightarrow q\bar{q}$, resulting in a corresponding modulation of $\Delta\varphi$. Such a modulation is much more pronounced in the $g \rightarrow q\bar{q}$ than in the $g \rightarrow gg$ branchings. As the $g \rightarrow gg$ process has a larger rate, and the two processes feature a modulation with opposite signs, the overall spin effect mostly cancels out. Therefore, the study of gluon spin correlations requires separating the different processes and selecting the $g \rightarrow q\bar{q}$ contribution in the second branching.

In this note, a measurement of the angle $\Delta\varphi$ of two successive parton radiations inside jets is reported, using pp collision data collected by the CMS experiment in 2022, corresponding to an integrated luminosity of 34.7 fb^{-1} . This is the first study of spin correlations induced by gluon polarization inside jets. Jets are reconstructed with the anti- k_T [16, 17] algorithm with distance parameter 0.8, denoted as AK8, and declustered using the Cambridge–Aachen (CA) algorithm [18]. The chain of declustered subjets are regarded as proxies of the parton radiation evolution. Two subsequent splittings are selected for each jet to calculate the angle $\Delta\varphi$ according to Eq. (1). A machine-learning (ML) approach is used to categorize events from different parton splittings and select those more likely to be $X \rightarrow Xg$, $g \rightarrow q\bar{q}$, where X can be either a gluon or a quark. The measured $\Delta\varphi$ distributions are compared to MC simulations

with spin correlations included (correlation-on) and neglected (correlation-off) to evaluate the size of the effect and examine the accuracy of the PS modeling.

The CMS apparatus [19, 20] is a multipurpose, nearly hermetic detector, designed to trigger [21–23] on and identify electrons, muons, photons, and (charged and neutral) hadrons [24–26]. A global “particle-flow” algorithm [27] aims to reconstruct all individual particles in an event, combining information provided by the all-silicon inner tracker and by the crystal electromagnetic and brass-scintillator hadron calorimeters, operating inside a 3.8 T superconducting solenoid, with data from the gas-ionization muon detectors embedded in the flux-return yoke outside the solenoid. The reconstructed particles are used to build τ leptons, jets, and missing transverse momentum [28–30].

Events of interest are selected using a two-tiered trigger system. The first level, composed of custom hardware processors, uses information from the calorimeters and muon detectors to select events at a rate of around 100 kHz within a fixed latency of 4 μ s [21]. The second level, known as the high-level trigger (HLT), consists of a farm of processors running a version of the full event reconstruction software optimized for fast processing, and reduces the event rate to a few kHz before data storage [22, 23].

Jet momentum is determined as the vectorial sum of all particle momenta in the jet, and is found from simulation to be, on average, within 5 to 10% of the true momentum over the whole p_T spectrum and detector acceptance. Additional pp interactions within the same or nearby bunch crossings (pileup) can contribute extra tracks and calorimetric energy depositions, increasing the apparent jet momentum. The pileup per particle identification algorithm (PUPPI) [31, 32] is used to mitigate these effect at the reconstructed particle level, making use of local shape information, event pileup properties, and tracking information. Charged particles identified to be originating from pileup vertices are discarded. The momenta of the neutral particles are rescaled according to their probability to originate from the primary interaction vertex deduced from the local shape variable, superseding the need for jet-based pileup corrections [31]. To obtain a sample of jet-enriched events with abundant parton splittings, jets of large size and high p_T are preferred. Events are selected at the HLT by requiring the presence of a jet with $p_T > 550$ GeV, corresponding to greater than 99.5% efficiency for jets with p_T above 700 GeV. Offline, the reconstructed AK8 jets are required to have $p_T > 700$ GeV and $|\eta| < 1.7$. The events must contain at least two jets separated in azimuth by two radians or more.

Simulations of dijet events are generated at leading order (LO) in perturbative QCD with PYTHIA 8.306 [33] with p_T -ordered shower and the CP5 tune [34], hereafter called PYTHIA 8, and with HERWIG 7.3.0 [35, 36] with angular shower and the CH3 tune [37], hereafter called HERWIG 7. Both MC generators use the NNPDF3.1 LO parton distribution functions (PDFs) [38]. The full detector response is simulated with GEANT4 [39].

The CA algorithm clusters particle pairs or subjets closest in angle sequentially, mimicking the angular ordering of LO parton splittings. In this analysis, charged and neutral particles with $p_T > 1$ GeV inside jets are used to reconstruct two successive parton splittings. The clustering of particles and subjets is repeated until all particles are recombined in a jet. Once the full jet is obtained, the backward clustering history gives the emission chain. For each jet, two successive branchings ($X \rightarrow 1 + 2$ and $2 \rightarrow 3 + 4$) are selected from the chain. The method proposed in Ref. [10] is followed, whereby subjet 1 and 2 (3 and 4) are the harder and softer subjets of the highest k_T splitting in the primary (secondary) Lund plane [40], with the k_T variable defined as

$$k_T = p_T^{\text{softer}} \cdot \Delta, \quad \Delta = \sqrt{(y_i - y_j)^2 + (\phi_i - \phi_j)^2}, \quad (2)$$

where p_T^{softer} is the transverse momentum of the softer subjet in the splitting, Δ represents the angular distance between two subjets i and j , and y_i and ϕ_i are the rapidity and azimuthal angle of subjet i . An extra $p_T > 130$ GeV requirement is applied for the subjet 2 to reduce the effect of softer radiations.

The selected subjets in the second branching, i.e. subjets 3 and 4, may originate from quarks or gluons. To identify the flavor of the reconstructed subjets, they are matched in the simulation to the subjets obtained via CA declustering of final state MC-truth partons with $p_T > 1$ GeV, after applying the selection used to identify the two branchings. The definition of the parton subjet flavor can be ambiguous since the multiple particles inside them do not necessarily originate from the same splitting, and can be affected by soft and collinear radiations. The Interleaved Flavour Neutralisation [41] prescription is applied, which provides infrared and collinear safe definitions of jet flavor at each stage of the clustering sequence. This is realized by a special treatment of the soft $q\bar{q}$ pairs. In ideal cases, the flavor of a subjet is either one of quarks or gluon, but the CA declustering might also include partons from other branchings. This happens in about 1% of the cases in the study, and results in a non-singlet flavor, such as $u\bar{d}$, referred to as the “rest” type. Reconstructed and parton subjets are matched if their distance is less than half of $\Delta R_{3,4}$, the separation between reconstructed subjets 3 and 4. Typically, $\Delta R_{3,4}$ is around 0.1. If either one of the subjets 3 and 4 are not matched to parton subjets, the pair is denoted as “unmatched”. With this approach, the fractions of qg , gg , $q\bar{q}$, “unmatched”, and “rest” types are found to be 7%, 24%, 4%, 65%, and 0.5%, respectively, in the HERWIG 7 simulation. The PYTHIA 8 results show a similar composition of 7%, 16%, 3%, 71%, and 1%. The unmatched events have typically more collinear and softer activity, with $\Delta R_{3,4} < 0.05$ and $p_T < 15$ GeV for subjet 4. Alternatively, one can obtain the flavor by matching subjet 2 and 3. In this case, the above fractions become 17%, 52%, 6%, 23%, and 0.7% for HERWIG 7; and 24%, 44%, 6%, 24%, and 2% for PYTHIA 8 simulations, respectively. The much smaller unmatched rate indicates that nonperturbative effects in the collinear and soft region, as well as limited detector resolution, have a large impact on subjet 4, and consequently smear the $\Delta\phi$ distribution.

Since the spin correlation effect is most pronounced in the $X \rightarrow Xg$, $g \rightarrow q\bar{q}$ topology, which account for a small fraction of all events, ML techniques are applied to improve their selection. The labeled subjets 3 and 4 in HERWIG 7 correlation-off model are used to train a deep neural network (DNN) to categorize reconstructed data in four classes: qg , gg , $q\bar{q}$, and unmatched. The “rest” type of events is negligible and not used in the training. Variables describing the internal structure of subjets 3 and 4, as well as the splitting features of the first and second branching are used for the training. In total, twelve variables are included:

- transverse momentum fraction of the subjet trailing in p_T in the first (z_1) and second (z_2) branching;
- angular distance between subjets 3 and 4 in the η - ϕ plane, $\Delta R_{3,4}$;
- k_T of the second $2 \rightarrow 3 + 4$ branching;
- number of charged and total particles in subjets 3 and 4;
- p_T -weighted width of the subjets 3 and 4, $\sigma = \frac{\sum_i p_T^i \Delta R^i}{\sum_i p_T^i}$, where p_T^i is the p_T of particle i inside a subjet, and ΔR^i the angular distance between particle i and the subjet axis in the η - ϕ plane;
- p_T dispersion of subjets 3 and 4, $p_T^D = \frac{\sqrt{\sum_i (p_T^i)^2}}{\sum_i p_T^i}$.

The DNN output consists of four scores, each representing the probability of a jet belonging to

one of the four classes. The distribution of the output score of a jet classified into the $q\bar{q}$ class in data and MC simulations, is shown in Fig. 1. The score differentiates well the various processes. The $q\bar{q}$ class distributes towards higher scores. The correlation-on and correlation-off models have identical score distributions, which ensures the independency of the DNN score and the spin correlation effects. Neither PYTHIA 8 nor HERWIG 7 simulations describe data well in the score distribution, due to the mismodeling of the input variables. A reweighting is applied to MC events to match the data. Since there are four output scores and they add up to unity for each jet, only three of them are independent. Therefore, a three-dimensional reweighting is applied to the $q\bar{q}$, $g\bar{g}$, and unmatched scores. After reweighting, the MC distributions of the input variables show a good agreement with data. Events are then classified into four categories according to the highest score among the four. About 50% of events are classified into the unmatched category, with a 42% purity of true unmatched events. Here, for the identification of the true flavor of the splitting, the matching of subjet 2 and 3 is used. The $g\bar{g}$ category takes up about 20% of the events with a purity of 75%. The $q\bar{q}$ category has 12% of the inclusive events and a 15% purity.

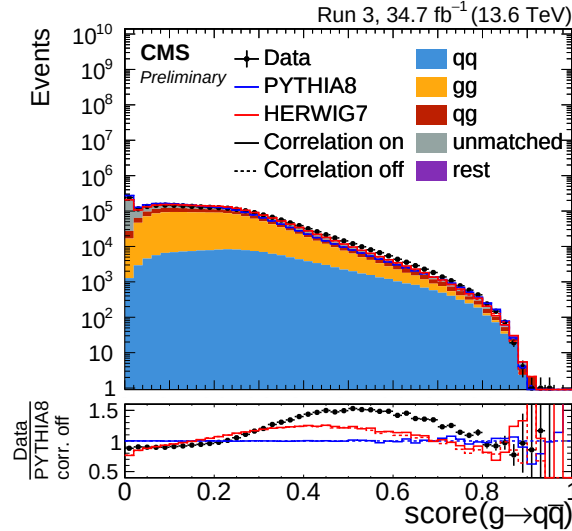


Figure 1: Distribution of the DNN output score for a jet identified in the $q\bar{q}$ class in the data and in the different MC simulations. Inclusive events are shown before categorization. The bottom panel shows the ratio of data to the PYTHIA 8 correlation-off prediction.

Figure 2 (left) shows the reconstructed $\Delta\phi$ distributions of inclusive events passing all selections. The data are compared to various MC predictions normalized to data. The difference between the correlation-on and correlation-off models is about 0.5%, due to the cancellation of polarization effects in the $g \rightarrow g\bar{g}$ and $g \rightarrow q\bar{q}$ processes and to the presence of a large portion of other event types without spin correlations. The PYTHIA 8 and HERWIG 7 models predict similar shapes. Data agree better with the correlation-on model, with a slightly larger modulation. The $\Delta\phi$ distributions in the unmatched, $q\bar{q}$, and $g\bar{g}$ categories can be found in Appendix A. They feature small spin correlations and serve as flavor-enriched regions for MC comparisons. The events in the $q\bar{q}$ category are used to study the impact of gluon polarization effects. An optimization as a function of the $q\bar{q}$ score selection is performed to maximize the significance against the correlation-off hypothesis. The test statistics q is defined as twice the log-likelihood ratio of the correlation-off and correlation-on hypotheses. Events with $\text{score}_{g \rightarrow q\bar{q}} > 0.6$ are thus selected, which enhances the purity of the $q\bar{q}$ class to 38% as well as the impact of gluon polarization effects. The differences between predictions with and without spin correlations are significantly larger in the $q\bar{q}$ category (Fig. 2, right) than in the

inclusive one (Fig. 2, left). A maximal 12% difference is seen in the $\Delta\phi$ distribution between the correlation-on and correlation-off PYTHIA 8 predictions, while HERWIG 7 shows stronger gluon-spin-induced correlations. Data favor the correlation-on models with a slightly smaller modulation than predicted.

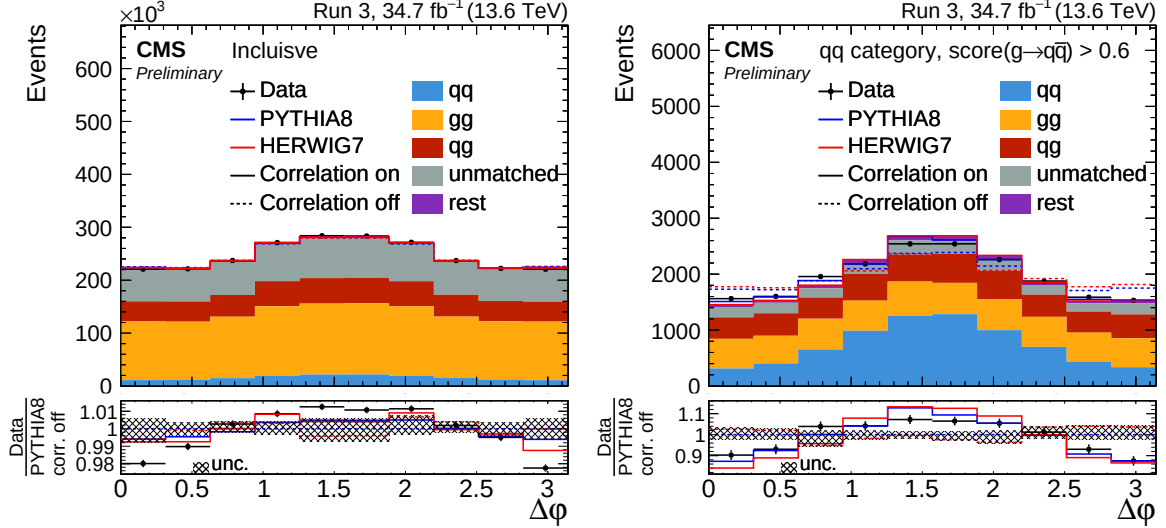


Figure 2: $\Delta\phi$ distributions in data and MC simulations for the inclusive sample (left), and in the $q\bar{q}$ category with $\text{score}_{g \rightarrow q\bar{q}} > 0.6$ (right). Systematic uncertainties are shown as grey bands. The bottom panels show the ratio of the data to the correlation-off PYTHIA 8 predictions.

Uncertainties arising from missing higher-order corrections in the hard scattering and parton shower are estimated with a seven-point method, where renormalization and factorization scales (μ_R and μ_F) are set independently to factors of 2, 1 and 0.5 times the relevant energy scale, with the ($\mu_R = 2$, $\mu_F = 0.5$) and ($\mu_R = 0.5$, $\mu_F = 2$) combinations removed from the total nine combinations. The uncertainty associated with the proton PDFs is evaluated using the envelope of variations obtained with the 100 NNPDF replica sets. The MC generators differ in their PS and hadronization models, as well as in the treatment of gluon-spin correlations, with PYTHIA 8 using an average effect and HERWIG 7 tracking explicitly the helicity evolution in the shower. The PYTHIA 8 predictions are used as a baseline, and their differences with respect to HERWIG 7 are treated as a single-sided shape uncertainty. This model difference together with the $\mu_{R,F}$ scale variations are the largest sources of theoretical systematic uncertainties, which amount to 0.5–3% and 0.5–1% in the $\Delta\phi$ distributions, respectively. The PDF uncertainty is smaller than 0.1%.

Experimental uncertainties include the energy scale and resolution of the jet, as well as the energy scale of the jet constituents. The latter are 3% for photons, 5% for neutral particles, and 1% for charged particles [42]. The event and subjet selections are repeated after varying the p_T of the jet and its constituents by their uncertainties. The track reconstruction efficiency uncertainty is 3%, due to mismodeling of charged-particle tracks in the dense core of the jets [29]. The pileup uncertainty is obtained by varying the inelastic pp cross section used to generate pileup interactions in the simulation. The uncertainty of the DNN score reweighting is assessed by varying the PYTHIA 8 MC flavor composition to that of HERWIG 7, via applying flavor-dependent scale factors to each event. The finite size of the simulated event samples introduces statistical fluctuations in the $\Delta\phi$ distribution. Therefore Poisson uncertainties per bin, at the scale of 1%, are added as systematic uncertainties. The largest experimental systematic uncertainties are the neutral particle energy scale and pileup, both have a 0.2–1% effect on the $\Delta\phi$ distribution.

The COMBINE tool [43] is used to obtain a statistical interpretation of the observed spin correlations in data. The two hypotheses, correlation-on and correlation-off, are tested against data using $\Delta\phi$ distributions of the events with $\text{score}_{g \rightarrow q\bar{q}} > 0.6$. Pseudodata is generated for the two hypotheses' shapes with the PYTHIA 8 simulation including systematic variations. The observed (expected) significance against correlation-off and correlation-on hypothesis are 6.8(8.1) and 1.9 standard deviations.

The measured $\Delta\phi$ distribution is unfolded to correct for instrumental effects. Exploiting its symmetry around $\pi/2$, the symmetrized $\pi/2 - |\Delta\phi - \pi/2|$ distribution is unfolded to reduce statistical uncertainties. The particle-level phase-space definition requires at least two AK8 jets with $p_T > 600$ GeV and $|\eta| < 1.7$. All particles with $p_T > 1$ GeV are used for CA declustering, and the hardest splittings in the primary and secondary Lund planes are selected. Subject 2 must have $p_T > 120$ GeV and a detector-level DNN is applied to particle-level subjects with $\text{score}_{g \rightarrow q\bar{q}} > 0.6$. Events passing both particle- and detector-level selections (with a $\approx 40\%$ overlap) define the migration matrix, with bin-to-bin migrations below 10%. Detector misreconstruction effects are removed from data, and particle-level inefficiencies are corrected using simulation. The modeling uncertainty is estimated from the difference between the results unfolded with PYTHIA 8 and HERWIG 7. Other uncertainties described above are propagated through the unfolding and added in quadrature in each bin. The leading systematic uncertainties are due to the modeling (1–7%) and tracking efficiency (1–3%). The unfolded distribution is shown in Fig. 3. Statistical and systematic uncertainties both range from 4% to 8%. The data agree with the PYTHIA 8 predictions within uncertainties, and show larger deviations with respect to the HERWIG 7 model.

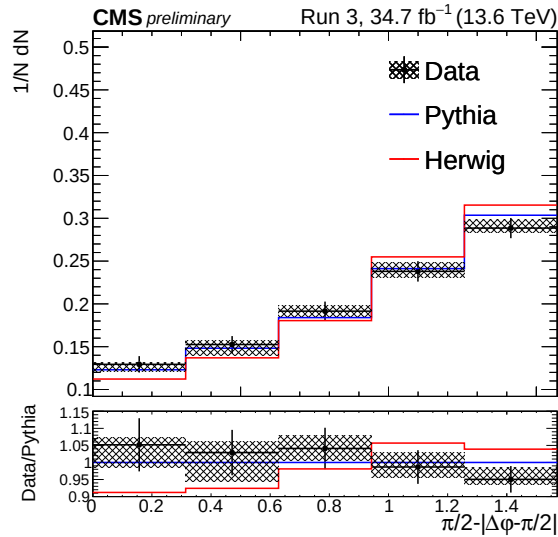


Figure 3: Unfolded $\Delta\phi$ distribution in events of the $q\bar{q}$ category with $\text{score}_{g \rightarrow q\bar{q}} > 0.6$ in data (dots) compared with PYTHIA 8 and HERWIG 7 predictions (histograms) passing the event selection described in the text. Systematic uncertainties are shown as grey bands. The bottom panel shows the ratio of the data to the correlation-on PYTHIA 8 predictions.

This note presents the first observation of spin correlations induced by the gluon polarization in soft and collinear parton emissions inside jets. The analysis uses proton-proton collision data at a center-of-mass energy of $\sqrt{s} = 13$ TeV, corresponding to an integrated luminosity of 34.7 fb^{-1} , collected with the CMS detector at the LHC in 2022. The details of the parton shower are studied using jets reconstructed with the anti- k_T algorithm and declustered with the Cambridge–Aachen algorithm. A new technique is developed to identify features in jet substructure and

to select intermediate gluon splittings into quark-antiquark pairs. An observable sensitive to the gluon polarization effect is compared with PYTHIA 8 and HERWIG 7 parton-shower predictions, with and without spin correlations. Data deviate from the null and spin-correlation models by 6.8 and 1.9 standard deviations, respectively, strongly disfavoring predictions that do not account for the gluon polarization. The spin correlations are apparent even in the presence of nonperturbative final-state effects, highlighting the importance of including the impact of gluon polarization in the event generators. The current models do not reproduce perfectly the shape of the angular modulation observed in the gluon-spin-sensitive distributions, and the unfolded measurement allows further improvements in the parton shower models. The study also presents the first attempt to experimentally identify light-quark and gluon subjects from intermediate parton emissions inside a jet, using collinear- and infrared-safe jet flavor definitions. The flavor tagging method can be further applied to the full splitting evolution inside a jet, so that the flavor dependence of parton splittings, e.g. in the Lund plane, can be studied in the future. This work opens the door to a broad use case in jet final states, such as exploring the spin correlation of gluon splittings in the search for Higgs boson decays into two gluons.

References

- [1] TASSO Collaboration, “Evidence for a spin one gluon in three jet events”, *Phys. Lett. B* **97** (1980) 453, doi:10.1016/0370-2693(80)90639-5.
- [2] PLUTO Collaboration, “A study of multi-jet events in e^+e^- annihilation”, *Phys. Lett. B* **97** (1980) 459, doi:10.1016/0370-2693(80)90640-1.
- [3] S. Moretti and W. J. Stirling, “Spin correlations in $e^+e^- \rightarrow$ four jets”, *Eur. Phys. J. C* **9** (1999) 81, doi:10.1007/s100529900016, arXiv:hep-ph/9808429.
- [4] ALEPH Collaboration, “A measurement of the QCD color factors and a limit on the light gluino”, *Z. Phys. C* **76** (1997) 1, doi:10.1007/s002880050522.
- [5] J. C. Collins, “Spin correlations in Monte Carlo event generators”, *Nucl. Phys. B* **304** (1988) 794, doi:10.1016/0550-3213(88)90654-2.
- [6] I. G. Knowles, “A linear algorithm for calculating spin correlations in hadronic collisions”, *Comput. Phys. Commun.* **58** (1990) 271, doi:10.1016/0010-4655(90)90063-7.
- [7] T. Sjöstrand et al., “An introduction to PYTHIA 8.2”, *Comput. Phys. Commun.* **191** (2015) 159, doi:10.1016/j.cpc.2015.01.024, arXiv:1410.3012.
- [8] Sherpa Collaboration, “Event generation with Sherpa 2.2”, *SciPost Phys.* **7** (2019) 034, doi:10.21468/SciPostPhys.7.3.034, arXiv:1905.09127.
- [9] P. Richardson and S. Webster, “Spin correlations in parton shower simulations”, *Eur. Phys. J. C* **80** (2020) 83, doi:10.1140/epjc/s10052-019-7429-5, arXiv:1807.01955.
- [10] A. Karlberg, G. P. Salam, L. Scyboz, and R. Verheyen, “Spin correlations in final-state parton showers and jet observables”, *Eur. Phys. J. C* **81** (2021) 681, doi:10.1140/epjc/s10052-021-09378-0, arXiv:2103.16526.
- [11] K. Hamilton et al., “Soft spin correlations in final-state parton showers”, *JHEP* **03** (2022) 193, doi:10.1007/JHEP03(2022)193, arXiv:2111.01161.

-
- [12] C. T. Preuss, “A partitioned dipole-antenna shower with improved transverse recoil”, *JHEP* **07** (2024) 161, doi:10.1007/JHEP07(2024)161, arXiv:2403.19452.
- [13] Z. Nagy and D. E. Soper, “Summations of large logarithms by parton showers”, *Phys. Rev. D* **104** (2021) 054049, doi:10.1103/PhysRevD.104.054049, arXiv:2011.04773.
- [14] J. R. Forshaw, J. Holguin, and S. Plätzer, “Parton branching at amplitude level”, *JHEP* **08** (2019) 145, doi:10.1007/JHEP08(2019)145, arXiv:1905.08686.
- [15] M. van Beekveld et al., “PanScales parton showers for hadron collisions: formulation and fixed-order studies”, *JHEP* **11** (2022) 019, doi:10.1007/JHEP11(2022)019, arXiv:2205.02237.
- [16] M. Cacciari, G. P. Salam, and G. Soyez, “The anti- k_T jet clustering algorithm”, *JHEP* **04** (2008) 063, doi:10.1088/1126-6708/2008/04/063, arXiv:0802.1189.
- [17] M. Cacciari, G. P. Salam, and G. Soyez, “FastJet user manual”, *Eur. Phys. J. C* **72** (2012) 1896, doi:10.1140/epjc/s10052-012-1896-2, arXiv:1111.6097.
- [18] Y. L. Dokshitzer, G. D. Leder, S. Moretti, and B. R. Webber, “Better jet clustering algorithms”, *JHEP* **08** (1997) 001, doi:10.1088/1126-6708/1997/08/001, arXiv:hep-ph/9707323.
- [19] CMS Collaboration, “The CMS experiment at the CERN LHC”, *JINST* **3** (2008) S08004, doi:10.1088/1748-0221/3/08/S08004.
- [20] CMS Collaboration, “Development of the CMS detector for the CERN LHC Run 3”, *JINST* **19** (2024) P05064, doi:10.1088/1748-0221/19/05/P05064, arXiv:2309.05466.
- [21] CMS Collaboration, “Performance of the CMS Level-1 trigger in proton-proton collisions at $\sqrt{s} = 13$ TeV”, *JINST* **15** (2020) P10017, doi:10.1088/1748-0221/15/10/P10017, arXiv:2006.10165.
- [22] CMS Collaboration, “The CMS trigger system”, *JINST* **12** (2017) P01020, doi:10.1088/1748-0221/12/01/P01020, arXiv:1609.02366.
- [23] CMS Collaboration, “Performance of the CMS high-level trigger during LHC Run 2”, *JINST* **19** (2024) P11021, doi:10.1088/1748-0221/19/11/P11021, arXiv:2410.17038.
- [24] CMS Collaboration, “Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC”, *JINST* **16** (2021) P05014, doi:10.1088/1748-0221/16/05/P05014, arXiv:2012.06888.
- [25] CMS Collaboration, “Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at $\sqrt{s} = 13$ TeV”, *JINST* **13** (2018) P06015, doi:10.1088/1748-0221/13/06/P06015, arXiv:1804.04528.
- [26] CMS Collaboration, “Description and performance of track and primary-vertex reconstruction with the CMS tracker”, *JINST* **9** (2014) P10009, doi:10.1088/1748-0221/9/10/P10009, arXiv:1405.6569.

- [27] CMS Collaboration, “Particle-flow reconstruction and global event description with the CMS detector”, *JINST* **12** (2017) P10003, doi:10.1088/1748-0221/12/10/P10003, arXiv:1706.04965.
- [28] CMS Collaboration, “Performance of reconstruction and identification of τ leptons decaying to hadrons and ν_τ in pp collisions at $\sqrt{s} = 13$ TeV”, *JINST* **13** (2018) P10005, doi:10.1088/1748-0221/13/10/P10005, arXiv:1809.02816.
- [29] CMS Collaboration, “Jet energy scale and resolution in the CMS experiment in pp collisions at 8 TeV”, *JINST* **12** (2017) P02014, doi:10.1088/1748-0221/12/02/P02014, arXiv:1607.03663.
- [30] CMS Collaboration, “Performance of missing transverse momentum reconstruction in proton-proton collisions at $\sqrt{s} = 13$ TeV using the CMS detector”, *JINST* **14** (2019) P07004, doi:10.1088/1748-0221/14/07/P07004, arXiv:1903.06078.
- [31] CMS Collaboration, “Pileup mitigation at CMS in 13 TeV data”, *JINST* **15** (2020) P09018, doi:10.1088/1748-0221/15/09/p09018, arXiv:2003.00503.
- [32] D. Bertolini, P. Harris, M. Low, and N. Tran, “Pileup per particle identification”, *JHEP* **10** (2014) 059, doi:10.1007/JHEP10(2014)059, arXiv:1407.6013.
- [33] C. Bierlich et al., “A comprehensive guide to the physics and usage of PYTHIA 8.3”, *SciPost Phys. Codeb.* **2022** (2022) 8, doi:10.21468/SciPostPhysCodeb.8, arXiv:2203.11601.
- [34] CMS Collaboration, “Extraction and validation of a new set of CMS PYTHIA8 tunes from underlying-event measurements”, *Eur. Phys. J. C* **80** (2020) 4, doi:10.1140/epjc/s10052-019-7499-4, arXiv:1903.12179.
- [35] M. Bahr et al., “Herwig++ physics and manual”, *Eur. Phys. J. C* **58** (2008) 639, doi:10.1140/epjc/s10052-008-0798-9, arXiv:0803.0883.
- [36] J. Bellm et al., “Herwig 7.0/Herwig++ 3.0 release note”, *Eur. Phys. J. C* **76** (2016) 196, doi:10.1140/epjc/s10052-016-4018-8, arXiv:1512.01178.
- [37] CMS Collaboration, “Development and validation of HERWIG7 tunes from cms underlying-event measurements”, *Eur. Phys. J. C* **81** (2021) 312, doi:10.1140/epjc/s10052-021-08949-5, arXiv:2011.03422.
- [38] NNPDF Collaboration, “Parton distributions for the LHC Run II”, *JHEP* **04** (2015) 040, doi:10.1007/JHEP04(2015)040, arXiv:1410.8849.
- [39] GEANT4 Collaboration, “GEANT4—a simulation toolkit”, *Nucl. Instrum. Meth. A* **506** (2003) 250, doi:10.1016/S0168-9002(03)01368-8.
- [40] F. A. Dreyer, G. P. Salam, and G. Soyez, “The Lund jet plane”, *JHEP* **12** (2018) 064, doi:10.1007/JHEP12(2018)064, arXiv:1807.04758.
- [41] F. Caola et al., “Flavored jets with exact anti- k_T kinematics and tests of infrared and collinear safety”, *Phys. Rev. D* **108** (2023) 094010, doi:10.1103/PhysRevD.108.094010, arXiv:2306.07314.
- [42] CMS Collaboration, “Measurement of energy correlators inside jets and determination of the strong coupling $\alpha_s(m_Z)$ ”, *Phys. Rev. Lett.* **133** (2024) 071903, doi:10.1103/PhysRevLett.133.071903, arXiv:2402.13864.

- [43] CMS Collaboration, “The CMS statistical analysis and combination tool: Combine”, *Comput. Softw. Big Sci.* **8** (2024) 19, doi:10.1007/s41781-024-00121-4, arXiv:2404.06614.

A Additional analysis figures

Figure A.1 shows the $\Delta\phi$ distribution measured in data compared to MC simulations in the unmatched (upper left), qg (upper right), and gg (lower) categories.

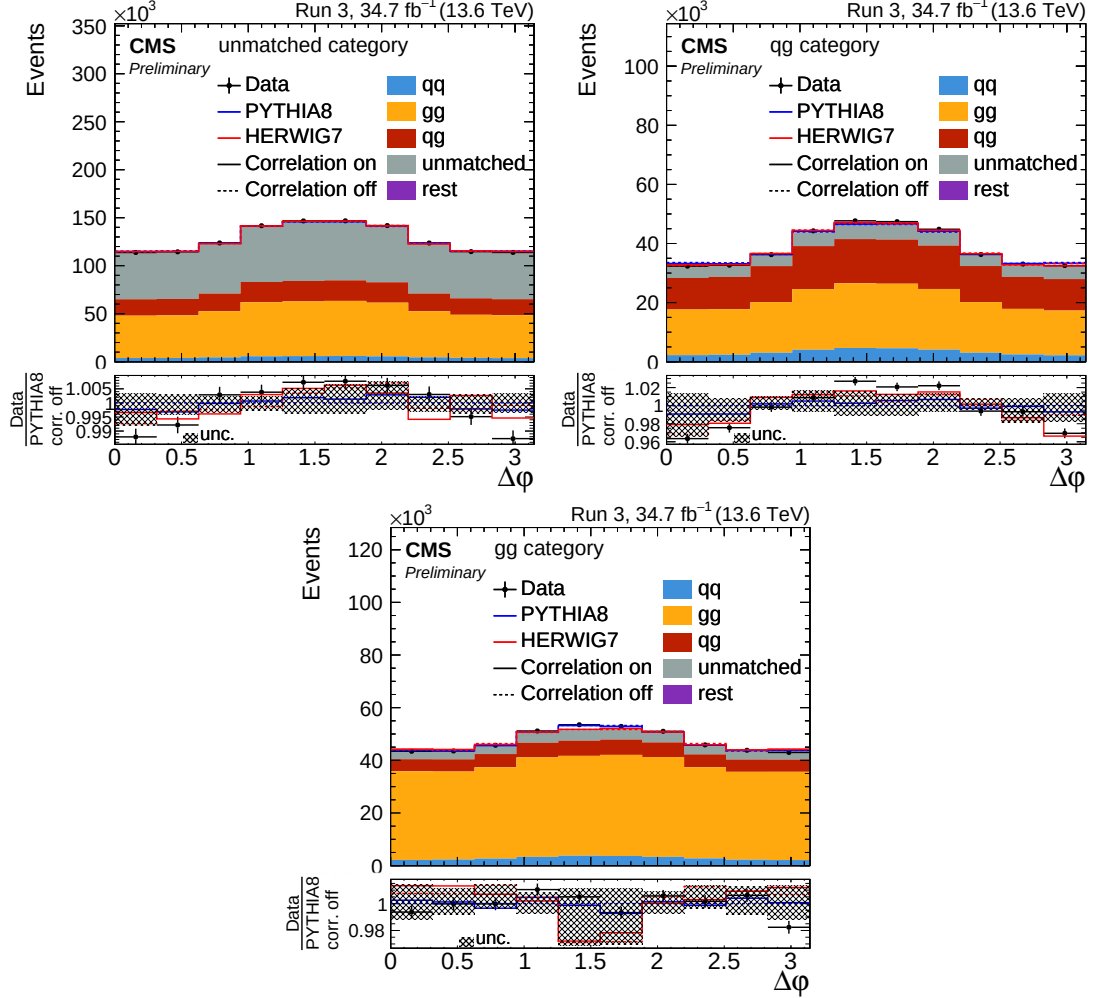


Figure A.1: Measured $\Delta\phi$ distributions in data (black dots) compared with the PYTHIA and HERWIG predictions (histograms) in the unmatched (upper left), qg (upper right), and gg (lower) categories. Systematic uncertainties are shown as grey bands. The bottom panels show the ratio of the data to the correlation-off PYTHIA 8 predictions.

Figure A.2 shows the $\Delta\phi$ distribution measured in data compared to MC simulations in the $q\bar{q}$ category with two different score selections.

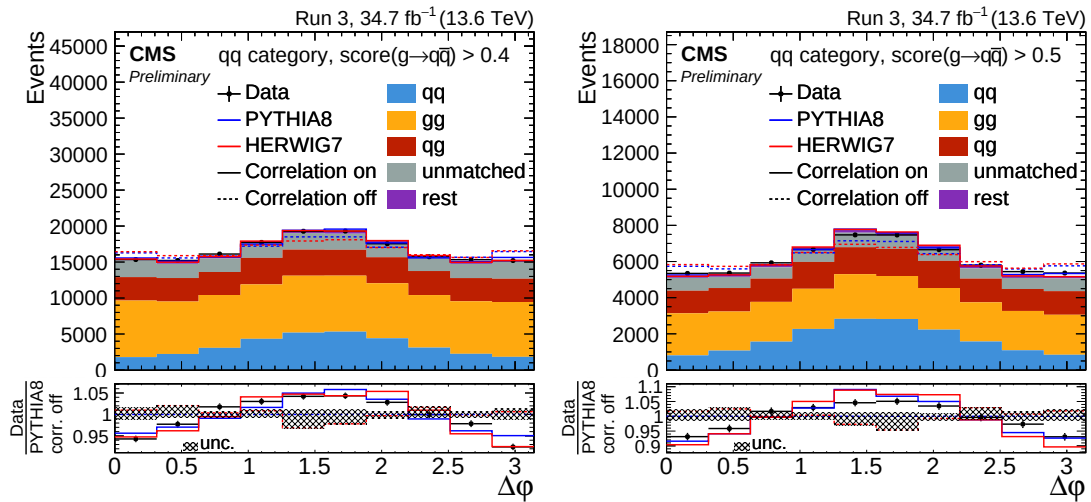


Figure A.2: Measured $\Delta\phi$ distributions measured in data (black dots) compared with the PYTHIA and HERWIG predictions (histograms) in the $q\bar{q}$ category with scores above 0.4 (left) and above 0.5 (right) selections. Systematic uncertainties are shown as grey bands. The bottom panels show the ratio of the data to the correlation-off PYTHIA 8 predictions.